Welcome to Design pattern

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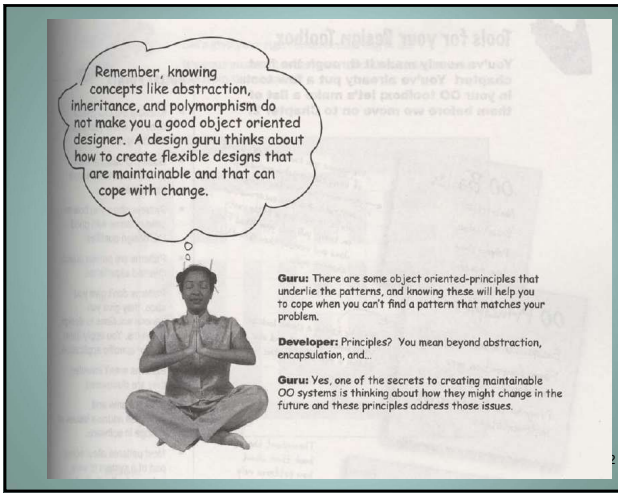

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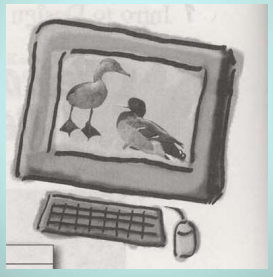
Joe works for a company that makes a highly successful duck pond simulation game, *SimUDuck*. The game can show a large variety of duck species swimming and making quacking sounds. The initial designers of the system used standard OO techniques and created one Duck superclass from which all other duck types inherit.



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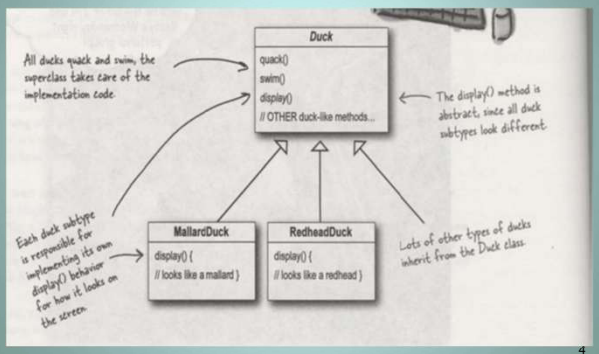
The OO design



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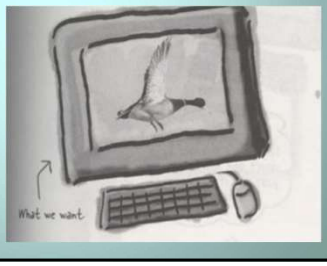
Remember ! Software change !

The executives decided that flying ducks is just what the simulator needs to blow away the other duck sim competitors. And of course Joe's manager told them it'll be no problem for Joe to just whip something up in a week. "After all", said Joe's boss, "he's an OO programmer... *how hard can it be?*"



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How are you going to change it?

You have 10 mins to think

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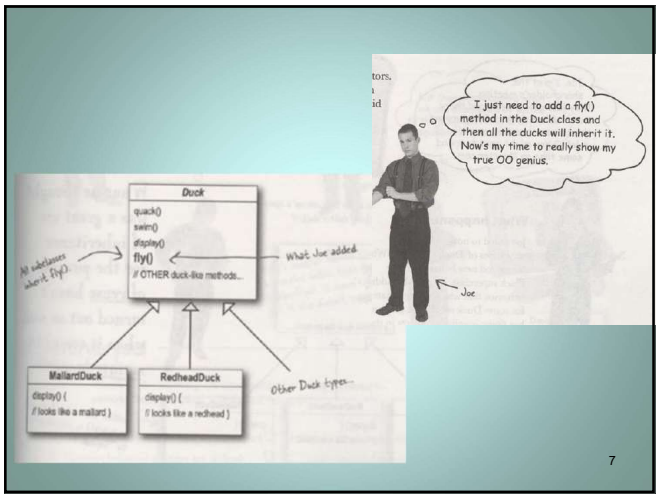
I just need to add a fly() method in the Duck class and then all the ducks will inherit it. Now's my time to really show my true OO genius.
Joe
Duck
quack()
swim()
display()
fly()
// OTHER duck-like methods...
All subclasses inherit fly()
What Joe added
MallardDuck
display() {
// looks like a mallard }
RedheadDuck
display() {
// looks like a redhead }
Other Duck types...
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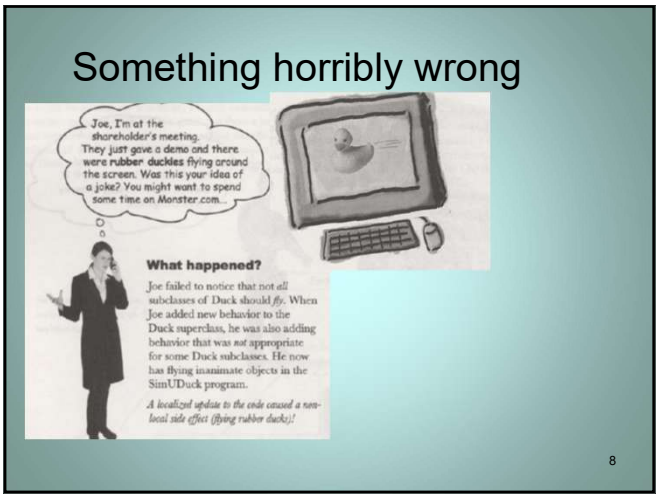
Something horribly wrong
Joe, I'm at the shareholder's meeting. They just gave a demo and there were rubber duckies flying around the screen. Was this your idea of a joke? You might want to spend some time on Monster.com...
What happened?
Joe failed to notice that not all subclasses of Duck should fly. When Joe added new behavior to the Duck superclass, he was also adding behavior that was not appropriate for some Duck subclasses. He now has flying inanimate objects in the SimUDuck program.
A localized update to the code caused a non-local side effect (flying rubber ducks)!
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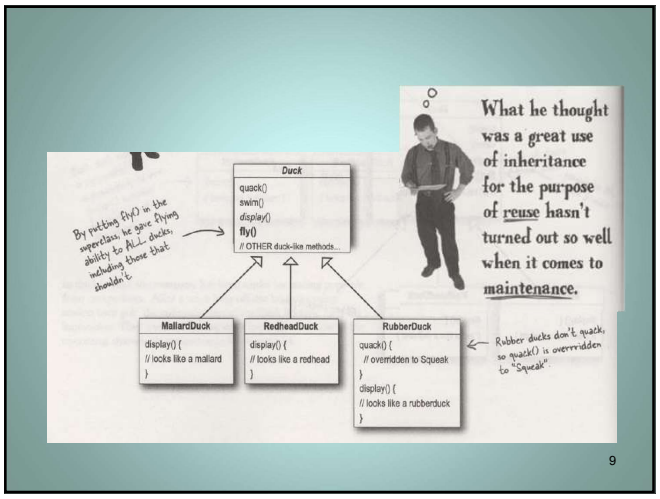
What he thought was a great use of inheritance for the purpose of reuse hasn't turned out so well when it comes to maintenance.
By putting fly() in the superclass, he gave flying ability to ALL ducks, including those that shouldn't.
Duck
quack()
swim()
display()
fly()
// OTHER duck-like methods...
MallardDuck
display() {
// looks like a mallard
}
RedheadDuck
display() {
// looks like a redhead
}
RubberDuck
quack() {
// overridden to Squeak
}
display() {
// looks like a rubberduck
}
Rubber ducks don't quack, so quack() is overridden to "Squeak"
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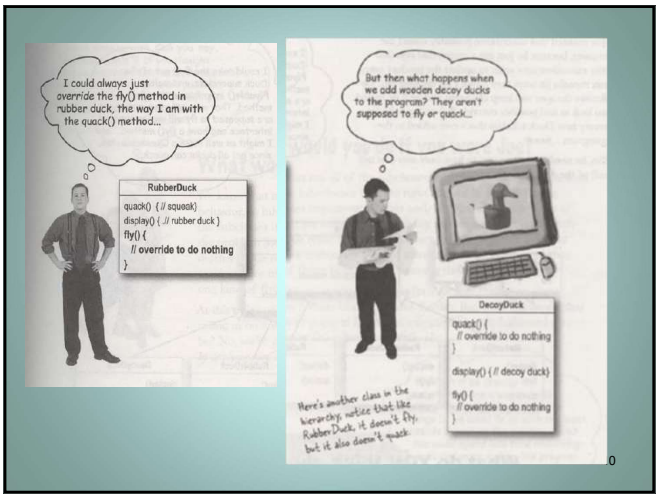
I could always just override the fly() method in rubber duck, the way I am with the quack() method...
RubberDuck
quack() { // squeak}
display() { // rubber duck }
fly() {
// override to do nothing
}
But then what happens when we add wooden decoy ducks to the program? They aren't supposed to fly or quack...
DecoyDuck
quack() {
// override to do nothing
}
display() { // decoy duck}
fly() {
// override to do nothing
}
There's another class in the hierarchy, notice that like RubberDuck, it doesn't fly, but it also doesn't quack.
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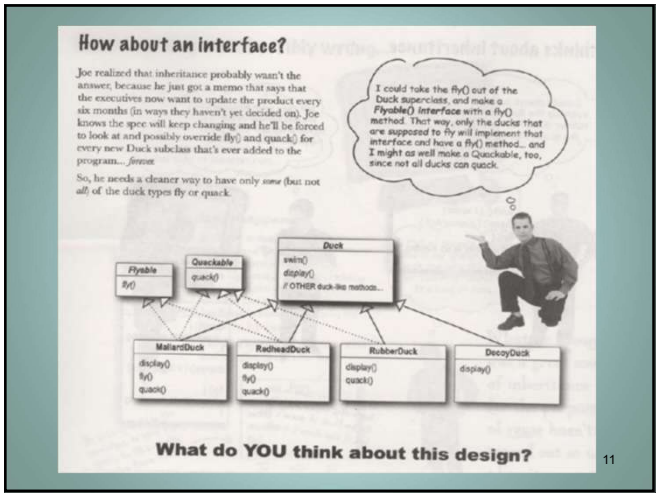
How about an interface?
Joe realized that inheritance probably wasn't the answer, because he just got a memo that says that the executives now want to update the product every six months (in ways they haven't yet decided on). Joe knows the spec will keep changing and he'll be forced to look at and possibly override fly() and quack() for every new Duck subclass that's ever added to the program... forever.
So, he needs a cleaner way to have only some (but not all) of the duck types fly or quack.
I could take the fly() out of the Duck superclass, and make a Flyable() interface with a fly() method. That way, only the ducks that are supposed to fly will implement that interface and have a fly() method... and I might as well make a Quackable, too, since not all ducks can quack.
Flyable
fly()
Quackable
quack()
Duck
swim()
display()
// OTHER duck-like methods...
MallardDuck
display()
fly()
quack()
RedheadDuck
display()
fly()
quack()
RubberDuck
display()
quack()
DecoyDuck
display()
What do YOU think about this design?
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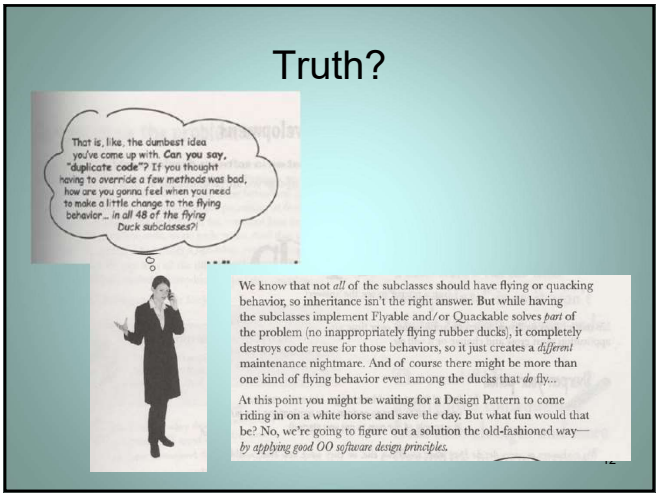
Truth?
That is, like, the dumbest idea you've come up with. Can you say, "duplicate code"? If you thought having to override a few methods was bad, how are you gonna feel when you need to make a little change to the flying behavior... in all 48 of the flying Duck subclasses?!
We know that not all of the subclasses should have flying or quacking behavior, so inheritance isn't the right answer. But while having the subclasses implement Flyable and/or Quackable solves part of the problem (no inappropriately flying rubber ducks), it completely destroys code reuse for those behaviors, so it just creates a different maintenance nightmare. And of course there might be more than one kind of flying behavior even among the ducks that do fly...
At this point you might be waiting for a Design Pattern to come riding in on a white horse and save the day. But what fun would that be? No, we're going to figure out a solution the old-fashioned way—by applying good OO software design principles.

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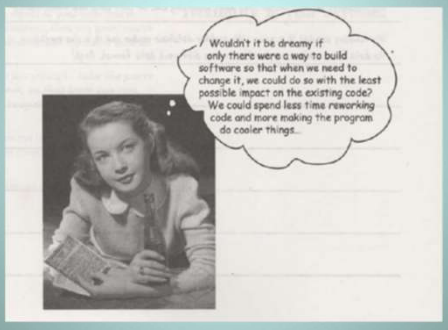
Recall the purpose of SE and OOAD



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The problem revisited

So we know using inheritance hasn't worked out very well, since the duck behavior keeps changing across the subclasses, and it's not appropriate for *all* subclasses to have those behaviors. The Flyable and Quackable interface sounded promising at first—only ducks that really do fly will be Flyable, etc.—except Java interfaces have no implementation code, so no code reuse. And that means that whenever you need to modify a behavior, you're forced to track down and change it in all the different subclasses where that behavior is defined, probably introducing *new* bugs along the way!

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Design principle

Take what varies and "encapsulate" it so it won't affect the rest of your code.

The result? Fewer unintended consequences from code changes and more flexibility in your systems!

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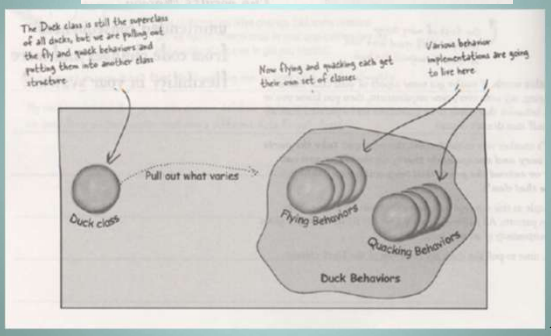
We know that fly() and quack() are the parts of the Duck class that vary across ducks.

To separate these behaviors from the Duck class, we'll pull both methods out of the Duck class and create a new set of classes to represent each behavior.



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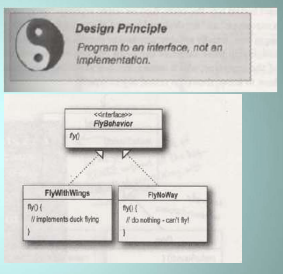
From now on, the Duck behaviors will live in a separate class—a class that implements a particular behavior interface.

That way, the Duck classes won't need to know any of the implementation details for their own behaviors.



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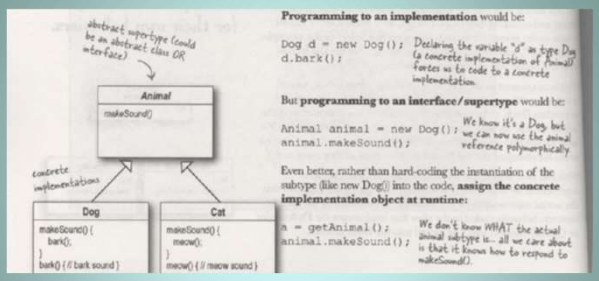

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Polymorphism

- What is polymorphism?

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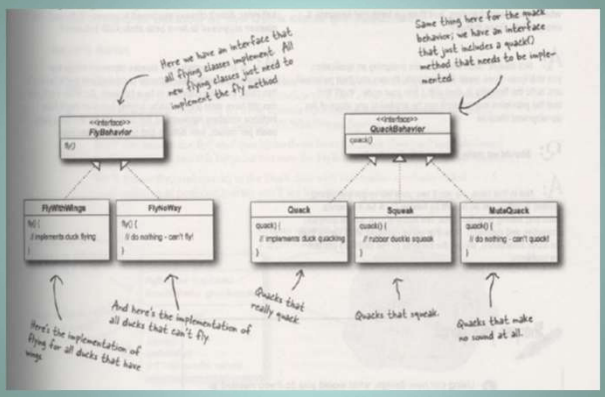

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With this design, other types of objects can reuse our fly and quack behaviors because these behaviors are no longer hidden away in our Duck classes!

And we can add new behaviors without modifying any of our existing behavior classes or touching any of the Duck classes that *use* flying behaviors.

So we get the benefit of REUSE without all the baggage that comes along with inheritance

Q: **It feels a little weird to have a class that's just a behavior. Aren't classes supposed to represent *things? Aren't classes supposed to have both state AND behavior?***

A: In an OO system, yes, classes represent things that generally have both state (instance variables) and methods. And in this case, the *thing* happens to be a behavior. But even a behavior can still have state and methods; a flying behavior might have instance variables representing the attributes for the flying (wing beats per minute, max altitude and speed, etc.) behavior.

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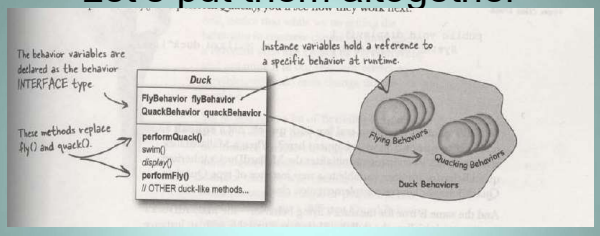
Let's put them altogether



Now we implement performQuack():

```
public class Duck {
    QuackBehavior quackBehavior;
    // more

    public void performQuack() {
        quackBehavior.quack();
    }
}
```

Each Duck has a reference to something that implements the QuackBehavior interface.

Rather than handling the quack behavior itself, the Duck object delegates that behavior to the object referenced by quackBehavior.

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To define a new kind of Duck

```
public class MallardDuck extends Duck {

    public MallardDuck() {
        quackBehavior = new Quack();
        flyBehavior = new FlyWithWings();
    }

    public void display() {
        System.out.println("I'm a real Mallard duck");
    }
}
```

A MallardDuck uses the Quack class to handle its quack, so when performQuack is called, the responsibility for the quack is delegated to the Quack object and we get a real quack.

And it uses FlyWithWings as its FlyBehavior type.

Remember, MallardDuck inherits the quackBehavior and flyBehavior instance variables from class Duck.

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The magic – you can have a duck whose behavior can change in runtime

So, we still have a lot of flexibility here, but we're doing a poor job of initializing the instance variables in a flexible way. But think about it, since the quackBehavior instance variable is an interface type, we could (through the magic of polymorphism) dynamically assign a different QuackBehavior implementation class at runtime.

Take a moment and think about how you would implement a duck so that its behavior could change at runtime. (You'll see the code that does this a few pages from now.)

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1 Type and compile the Duck class below (Duck.java), and the MallardDuck class from two pages back (MallardDuck.java).

```
public abstract class Duck {

    FlyBehavior flyBehavior;
    QuackBehavior quackBehavior;
    public Duck() {
    }

    public abstract void display();

    public void performFly() {
        flyBehavior.fly();
    }

    public void performQuack() {
        quackBehavior.quack();
    }

    public void swim() {
        System.out.println("All ducks float, even decoys!");
    }
}
```

Declare two reference variables for the behavior interface types. All duck subclasses (in the same package) inherit these.

Delegate to the behavior class.

2 Type and compile the FlyBehavior interface (FlyBehavior.java) and the two behavior implementation classes (FlyWithWings.java and FlyNoWay.java).

```
public interface FlyBehavior {
    public void fly();
}
```

The interface that all flying behavior classes implement.

```
public class FlyWithWings implements FlyBehavior {
    public void fly() {
        System.out.println("I'm flying!!");
    }
}
```

Flying behavior implementation for ducks that DO fly.

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4 Type and compile the test class (MiniDuckSimulator.java).

```
public class MiniDuckSimulator {
    public static void main(String[] args) {
        Duck mallard = new MallardDuck();
        mallard.performQuack();
        mallard.performFly();
    }
}
```

This calls the MallardDuck's inherited performQuack() method, which then delegates to the object's QuackBehavior (i.e. calls quack() on the duck's inherited quackBehavior reference).

Then we do the same thing with MallardDuck's inherited performFly() method.

5 Run the code!

```
% java MiniDuckSimulator
Quack
I'm flying!!
```

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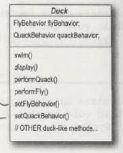
Setting behavior dynamically

What a shame to have all this dynamic talent built into our ducks and not be using it! Imagine you want to set the duck's behavior type through a setter method on the duck subclass, rather than by instantiating it in the duck's constructor.

1 Add two new methods to the Duck class:

```
public void setFlyBehavior(FlyBehavior fb) {
    flyBehavior = fb;
}

public void setQuackBehavior(QuackBehavior qb) {
    quackBehavior = qb;
}
```



We can call these methods anytime we want to change the behavior of a duck on the fly.

editor note: gratuitous pun - fix

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2 Make a new Duck type (ModelDuck.java).

```
public class ModelDuck extends Duck {
    public ModelDuck() {
        flyBehavior = new FlyNoWay();
        quackBehavior = new Quack();
    }

    public void display() {
        System.out.println("I'm a model duck");
    }
}
```

Our model duck begins life grounded... without a way to fly.

3 Make a new FlyBehavior type (FlyRocketPowered.java).

```
public class FlyRocketPowered implements FlyBehavior {
    public void fly() {
        System.out.println("I'm flying with a rocket!");
    }
}
```

That's okay, we're creating a rocket powered flying behavior.

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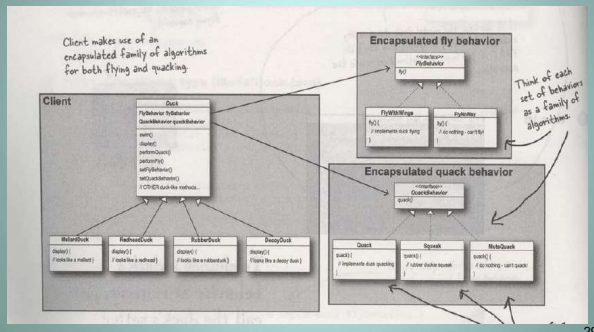
An entire solution

Pay careful attention to the *relationships* between the classes. In fact, grab your pen and write the appropriate relationship (IS-A, HAS-A and IMPLEMENTS) on each arrow in the class diagram.



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Design Principle

Favor composition over inheritance.

HAS-A can be better than IS-A

The HAS-A relationship is an interesting one: each duck has a FlyBehavior and a QuackBehavior to which it delegates flying and quacking.

When you put two classes together like this you're using ***composition***. Instead of *inheriting* their behavior, the ducks get their behavior by being *composed* with the right behavior object.

This is an important technique; in fact, we've been using our third design principle:

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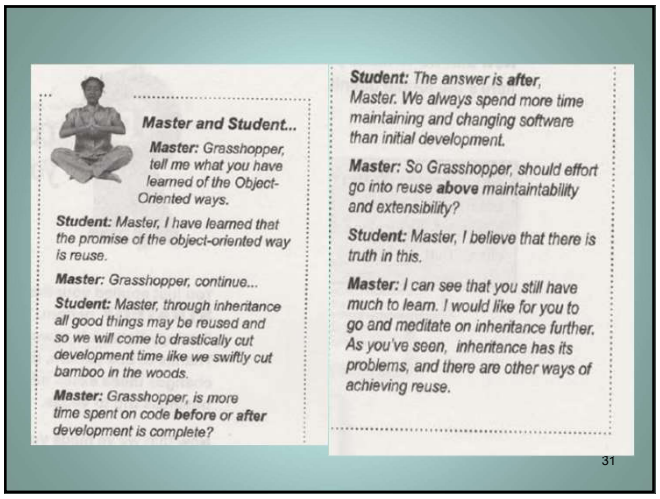

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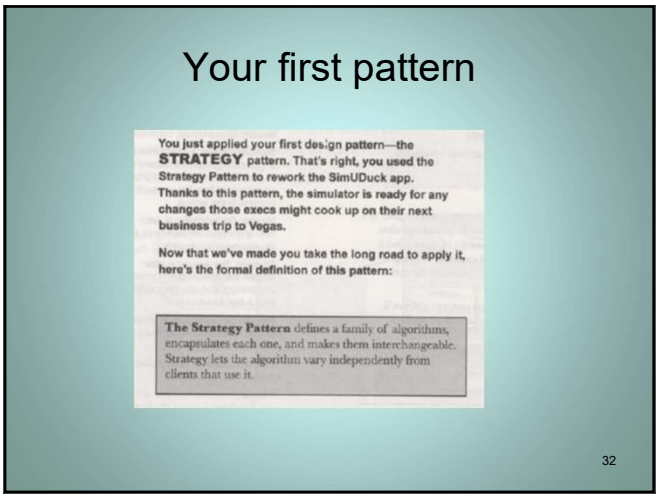

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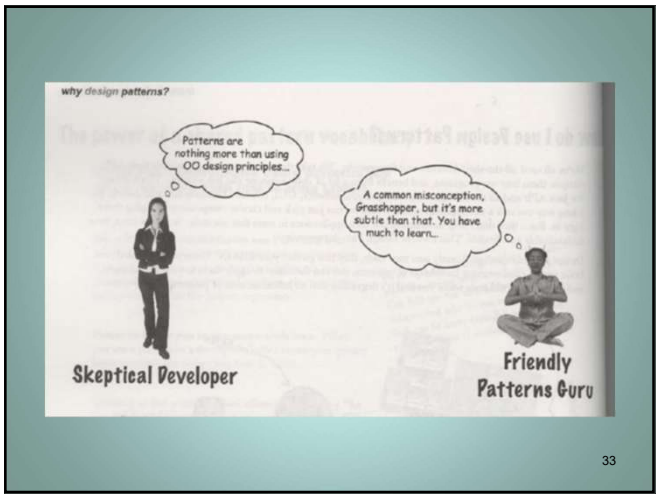

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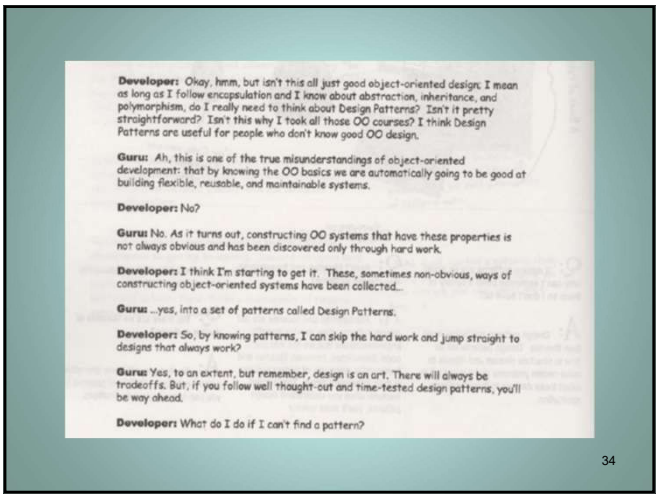

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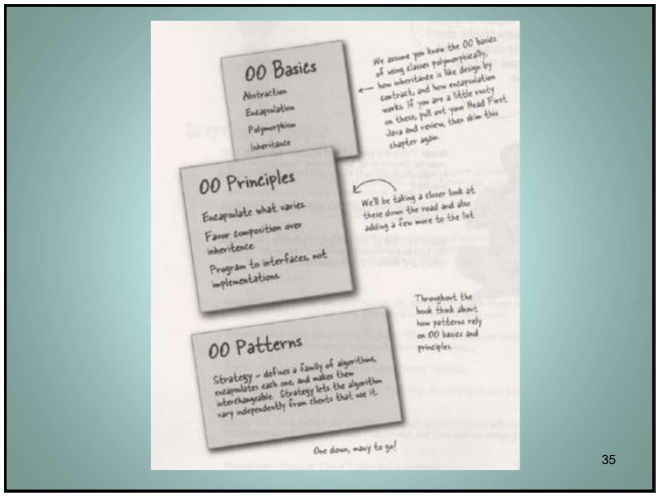

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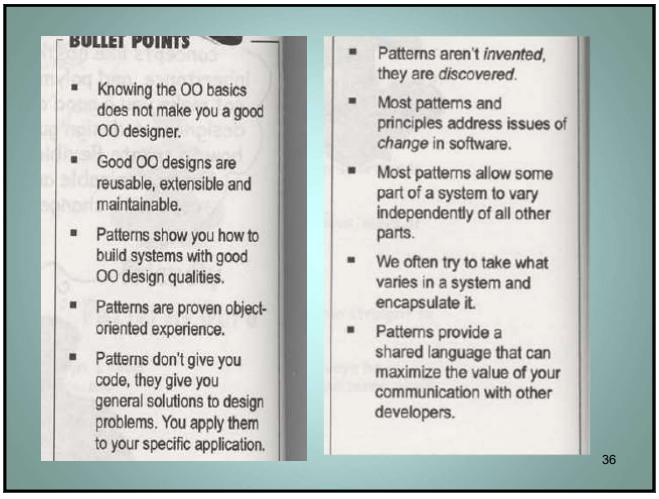

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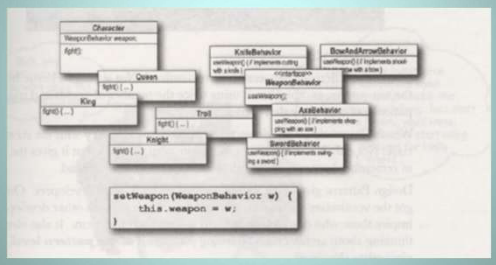
Below you'll find a mess of classes and interfaces for an action adventure game. You'll find classes for game characters along with classes for weapon behaviors the characters can use in the game. Each character can make use of one weapon at a time, but can change weapons at any time during the game. Your job is to sort it all out...

(Answers are at the end of the chapter.)



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